

SHARED TRANSCRIPTIONAL PROGRAMS ASSOCIATED WITH METASTASIS IN GASTROENTEROPANCREATIC NEUROENDOCRINE TUMORS

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ABSTRACT

Neuroendocrine tumors (NETs) are a heterogeneous class of malignancies whose clinical outcomes are driven by metastatic progression. Among gastroenteropancreatic neuroendocrine tumors (GEP-NETs), pancreatic neuroendocrine tumors (PanNETs) and small intestinal neuroendocrine tumors (SI-NETs) arise from distinct anatomical sites yet share common neuroendocrine features. While prior transcriptomic studies have examined metastatic mechanisms within individual NET subtypes, the extent to which metastatic progression is governed by conserved transcriptional programs across tissues remains unclear. In this project, RNA sequencing (RNA-seq) data were analyzed to compare metastatic and primary tumors within PanNET and SI-NET cohorts. Differential expression analysis, cross-tissue overlap assessment, pathway enrichment, and signature-based validation were performed. This analysis revealed a large, directionally consistent set of metastasis-associated genes enriched for metabolic processes. A shared metastasis signature score was significantly elevated in metastatic samples across both tumor types, supporting the existence of a conserved transcriptional program underlying neuroendocrine tumor metastasis.

Keywords: neuroendocrine tumors, RNA-seq, metastasis, pathway enrichment, gene signatures, bioinformatics

1. Introduction

Neuroendocrine tumors (NETs) are a diverse group of malignancies that arise from neuroendocrine cells, which possess both neuronal and endocrine characteristics. These tumors can occur throughout the body, but a major subset originates within the gastroenteropancreatic system, collectively referred to as gastroenteropancreatic neuroendocrine tumors (GEP-NETs). Among these, pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs) represent two of the most clinically relevant and biologically distinct subtypes [1,2]. Despite differences in tissue origin and molecular features, both PanNETs and SI-NETs share a common

clinical challenge: disease progression is largely driven by metastasis, which is associated with significantly worse patient outcomes [1,2].

Metastasis is a complex, multistep process that involves widespread changes in cellular behavior rather than the dysregulation of a single gene or pathway. In many cancers, metastatic progression reflects a shift in overall cellular state, encompassing alterations in metabolism, stress response, and interactions with the surrounding microenvironment [3,4]. Neuroendocrine tumors are particularly heterogeneous, both between patients and within individual tumors, making it difficult to identify consistent molecular drivers of metastasis using gene-by-gene approaches alone [5]. As a result, there is growing interest in

identifying broader transcriptional programs that characterize metastatic behavior across tumor subtypes.

RNA sequencing (RNA-seq) has become a central tool for studying genome-wide transcriptional changes in cancer. By quantifying expression levels across thousands of genes simultaneously, RNA-seq enables the identification of differentially expressed genes as well as higher-order biological patterns through pathway- and gene-set-based analyses [6]. Publicly available repositories such as the Gene Expression Omnibus (GEO) provide access to large, well-annotated RNA-seq datasets, enabling reproducible and comparative analyses across tumor types [7]. In this project, RNA-seq data were leveraged to study transcriptional differences between primary and metastatic neuroendocrine tumors in a systematic and integrative manner.

While many transcriptomic studies of neuroendocrine tumors focus on a single tissue type, cross-tissue comparative approaches offer an opportunity to distinguish tissue-specific effects from conserved biological programs. Large-scale cancer profiling studies have shown that tissue of origin is often a dominant source of transcriptional variation, underscoring the importance of accounting for tissue context when studying cancer biology [8]. PanNETs and SI-NETs arise from different anatomical locations yet share a neuroendocrine lineage and similar patterns of metastatic spread, making them well suited for comparative analysis. Identifying transcriptional changes that are consistently associated with metastasis across both tumor types may therefore reveal fundamental mechanisms of neuroendocrine tumor progression that are not restricted to a single tissue context.

In this project, differential expression analysis was performed separately for PanNET and SI-NET samples to compare metastatic tumors with primary tumors. Differentially expressed genes were identified within each tissue type, and overlap analysis was used to determine the extent to which metastasis-associated transcriptional changes were shared across tissues. Both the presence of shared genes and the consistency of their direction of regulation were evaluated, allowing for the

identification of a conserved metastasis-associated gene set.

To move beyond individual gene lists, pathway enrichment analysis was conducted on genes consistently upregulated in metastatic tumors across both PanNETs and SI-NETs. Gene Ontology (GO) biological process enrichment was used to identify common functional themes, revealing coordinated biological processes associated with metastatic progression [9]. In addition, a shared metastasis gene signature was constructed from these conserved genes, and a signature score was calculated for each sample using normalized RNA-seq expression data. This signature-based approach enabled validation of the shared transcriptional program at the sample level and allowed direct comparison between primary and metastatic tumors across tissues.

Overall, this project aimed to identify and characterize conserved transcriptional programs associated with metastasis in gastroenteropancreatic neuroendocrine tumors. By integrating differential expression analysis, cross-tissue overlap assessment, pathway enrichment, and signature-based validation, this study provides insight into shared molecular features of neuroendocrine tumor metastasis and demonstrates the utility of program-level transcriptomic analysis for understanding complex cancer phenotypes.

2. Methods

2.1 Data Source and Cohort Definition

All data used in this study were obtained from a publicly available RNA-sequencing dataset in the Gene Expression Omnibus (GEO) under accession GSE98894 [10]. This dataset contains RNA-seq profiles from gastroenteropancreatic neuroendocrine tumors. Raw gene-level read counts were used so that all normalization and statistical modeling could be performed internally by DESeq2.

Sample metadata were extracted from the GEO series matrix file and used to annotate tumors by anatomical site and disease status. Tumors originating from the pancreas were classified as pancreatic neuroendocrine tumors (PanNETs), while tumors originating from the small intestine or ileum were classified as small-intestinal

neuroendocrine tumors (SI-NETs). Rectal neuroendocrine tumors were excluded from downstream analysis due to their smaller sample size and distinct biological characteristics. After filtering, the final cohort consisted of 194 samples, including both primary tumors and metastases.

2.2 Preprocessing and DESeq2 Dataset Construction

Raw RNA-sequencing read counts were used as the starting point for all analyses. Each value in the count matrix represents the number of sequencing reads aligned to a given gene in each tumor sample. Raw counts were used intentionally, as DESeq2 performs its own normalization and statistical modeling and requires untransformed count data as input [11].

Before analysis, genes with extremely low expression across the dataset were removed. Specifically, genes were required to have a total read count greater than 10 across all samples. Genes with very few reads across the entire cohort are dominated by technical noise and cannot be reliably tested for differential expression. Removing these genes improves statistical power while preserving biologically meaningful expression patterns.

Differential expression analysis was performed using DESeq2, a method specifically developed for RNA-seq count data. DESeq2 corrects for differences in sequencing depth between samples by estimating sample-specific size factors, allowing expression levels to be compared fairly across tumors. In addition, DESeq2 models gene-level expression variability using a negative binomial distribution [11], which is well suited to RNA-seq data due to its ability to handle overdispersion and biological variability.

An initial DESeq2 dataset was constructed using a design formula that included both anatomical site and disease status. This design allowed global transcriptional patterns to be explored while accounting for large baseline differences in gene expression between pancreatic and small-intestinal tumors. This step was used for exploratory analysis and quality control rather than for final hypothesis testing.

To specifically identify gene expression changes associated with metastatic progression, samples were then analyzed in a tissue-stratified manner. The dataset was split into pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs), and differential expression analysis was performed independently within each tissue type. In these tissue-specific analyses, disease status was used as the sole predictor variable, with primary tumors designated as the reference condition. This approach ensured that identified differentially expressed genes reflected metastasis-associated transcriptional changes within each tissue rather than differences driven by tissue of origin.

2.3 Variance Stabilization and Principal Component Analysis

RNA-seq count data exhibit a strong relationship between mean expression level and variance, with highly expressed genes showing greater variability than lowly expressed genes. To reduce this dependency and make gene expression values more comparable across genes and samples, a variance-stabilizing transformation (VST) was applied to the count data using DESeq2 [11]. This transformation preserves relative expression patterns while producing values that are more suitable for distance-based analyses.

Principal component analysis (PCA) was used as an exploratory tool to summarize global gene expression patterns and to visualize relationships among tumor samples. PCA reduces the dimensionality of the dataset by identifying new axes, called principal components, that capture the largest sources of variation in gene expression across samples. Each sample is then represented as a point in a low-dimensional space defined by these components [12].

PCA was first performed using all samples to evaluate the dominant sources of transcriptomic variation across the dataset. This analysis was used to assess whether samples clustered according to known biological variables, such as tissue of origin and metastatic status, and to identify any potential outlier samples that might indicate technical artifacts or annotation errors.

Because tissue of origin represents a major source of gene expression variation in

gastroenteropancreatic neuroendocrine tumors, PCA was subsequently performed separately within pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). This tissue-specific PCA approach allowed variation associated with metastatic status to be examined without the confounding influence of large baseline differences between anatomical sites.

2.4 Differential Expression Analysis

Differential expression analysis was performed to identify genes whose expression levels differed between metastatic and primary tumors within each tissue type. Because pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs) have distinct baseline expression profiles, analyses were conducted separately for each tissue to ensure that observed differences reflected metastatic progression rather than tissue-of-origin effects.

Within each tissue-specific dataset, metastatic tumors were compared directly to primary tumors using DESeq2. Primary tumors were designated as the reference condition, so positive log₂ fold change values indicate higher expression in metastatic tumors, while negative values indicate higher expression in primary tumors [11].

DESeq2 estimates gene-wise dispersion and fits a negative binomial model for each gene to test whether observed expression differences are greater than expected by chance. To improve the stability and interpretability of effect size estimates, log₂ fold changes were shrunk using the apeglm method. Shrinkage reduces exaggerated fold changes for low-count or highly variable genes while preserving large, well-supported expression differences [13].

Statistical significance was assessed using adjusted p-values to control the false discovery rate. Genes with an adjusted p-value less than 0.05 were considered significantly differentially expressed. DESeq2's built-in independent filtering was used to exclude genes with insufficient expression levels from hypothesis testing [11], improving statistical power without introducing manual bias.

2.5 Cross-Tissue Overlap and Directionality Analysis

To determine whether metastatic gene expression changes were shared across tumor types, the results of the tissue-specific differential expression analyses were compared between pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). This comparison was designed to identify genes whose expression changes were associated with metastasis in both tissues rather than being specific to a single anatomical site.

For each tissue, genes that met the significance threshold for differential expression were collected into tissue-specific gene lists. These lists were then compared using set intersection to identify genes that were significantly differentially expressed in both PanNETs and SI-NETs.

For genes present in both lists, the direction of differential expression was examined by comparing the sign of the log₂ fold change in each tissue. Genes were considered directionally consistent if they were either upregulated in metastatic tumors in both tissues or downregulated in metastatic tumors in both tissues. Genes showing opposite directions of change between tissues were classified as directionally inconsistent.

This analysis allowed shared metastasis-associated genes to be distinguished from genes whose expression changes were tissue-specific, providing a focused set of conserved transcriptional changes associated with metastatic progression across gastroenteropancreatic neuroendocrine tumors.

2.6 Pathway Enrichment Analysis

After identifying genes that were consistently associated with metastasis across both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs), pathway enrichment analysis was performed to understand the biological processes represented by these shared genes. Rather than focusing on individual genes in isolation, this analysis aimed to determine whether groups of functionally related genes were overrepresented among metastasis-associated transcriptional changes.

Genes that were significantly upregulated in metastatic tumors in both tissue types were selected for pathway analysis. These genes represent a conserved set of transcriptional changes associated

with metastatic progression across gastroenteropancreatic neuroendocrine tumors.

Gene Ontology (GO) biological process enrichment analysis was conducted using the clusterProfiler package [14,15]. This approach evaluates whether specific biological processes occur more frequently in the shared metastatic gene set than would be expected by chance, given the background of all expressed genes. Enrichment significance was assessed using Benjamini–Hochberg correction to control for multiple hypothesis testing. [15]

GO terms with adjusted p-values less than 0.05 were considered significantly enriched. Enrichment results were summarized and visualized using dot plots to highlight dominant biological processes associated with the shared metastasis-associated transcriptional program.

2.7 Construction of a Shared Metastasis Gene Signature

To quantify the activity of conserved metastasis-associated transcriptional programs at the sample level, a shared metastasis gene signature was constructed. This signature was based on genes that were consistently upregulated in metastatic tumors in both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs), ensuring that the signature reflected shared biological changes rather than tissue-specific effects.

For each gene in the shared signature, variance-stabilized expression values were standardized across samples using z-score normalization. This step places all genes on a common scale, allowing genes with different baseline expression levels to contribute equally to the signature. Standardization ensures that the signature score reflects coordinated expression patterns across genes rather than being dominated by a small number of highly expressed genes.

A single signature score was calculated for each tumor sample by averaging the standardized expression values across all genes in the shared signature. This score represents the overall level of

activation of the shared metastasis-associated transcriptional program within each sample.

Signature scores were compared between primary and metastatic tumors to assess whether the shared gene program was more strongly expressed in metastatic disease. Statistical significance was evaluated using Welch’s two-sample t-test, which does not assume equal variance between groups. To examine tissue-specific behavior, signature scores were also evaluated separately within PanNET and SI-NET cohorts.

2.8 Software, Visualization, and Statistical Analysis

All analyses were performed using the R statistical computing environment [16]. Data manipulation and analysis workflows were implemented using tidyverse-compatible packages [17]. Differential expression analysis and variance-stabilizing transformation were performed using DESeq2, and pathway enrichment analysis was conducted using clusterProfiler with Gene Ontology annotations.

Data visualizations were generated using ggplot2 [18] and pheatmap [19]. Principal component analysis plots, volcano plots, heatmaps, pathway enrichment dot plots, and signature score visualizations were produced using standardized plotting workflows to ensure consistency across figures.

Unless otherwise stated, statistical tests were two-sided. Adjusted p-values were used where applicable to control for multiple hypothesis testing.

3. Results

3.1 Global transcriptomic structure of gastroenteropancreatic neuroendocrine tumors

To assess the dominant sources of transcriptional variation across gastroenteropancreatic neuroendocrine tumors, principal component analysis (PCA) was performed on variance-stabilized RNA-seq expression data from all 194 samples. This analysis served as an initial exploratory step to evaluate overall dataset structure, biological coherence, and the relative

contributions of tissue of origin and metastatic status to global gene expression patterns.

PCA revealed a clear separation by tissue of origin along the first principal component (PC1), which accounted for 28% of the total variance (Figure 1). Pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs) formed largely distinct clusters with minimal overlap, indicating that anatomical site represents the dominant driver of transcriptomic variation across the dataset. This strong separation is biologically intuitive and confirms that samples are correctly annotated and free from major technical artifacts.

In contrast, metastatic status did not produce a strong global separation across all samples. Primary

and metastatic tumors largely overlapped within each tissue cluster, although metastatic samples tended to display increased dispersion. This pattern suggests that metastatic progression does not override tissue-specific transcriptional identity, but instead introduces additional, context-dependent expression changes layered on top of the underlying tissue-specific program.

Together, these results demonstrate that tissue identity is the primary determinant of global gene expression structure in gastroenteropancreatic neuroendocrine tumors. This observation motivated subsequent tissue-stratified analyses to evaluate metastasis-associated transcriptional changes within PanNET and SI-NET cohorts independently.

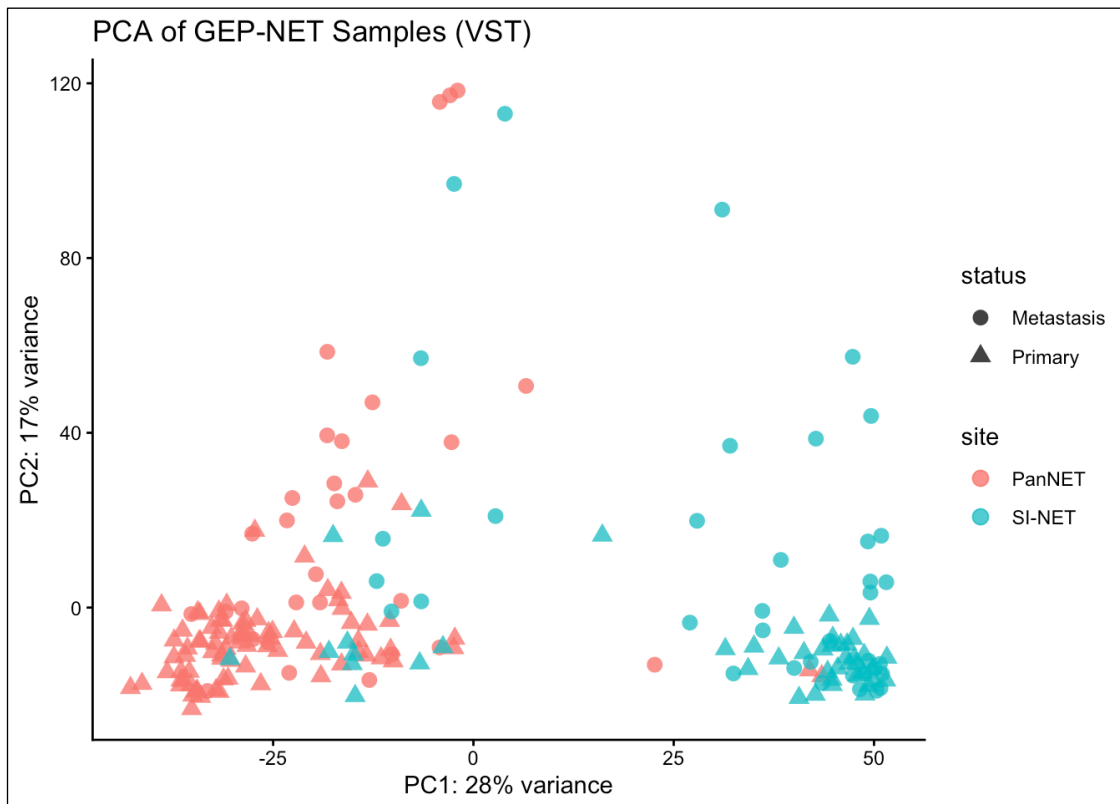


Figure 1. Global transcriptomic structure of gastroenteropancreatic neuroendocrine tumors. Principal component analysis (PCA) of variance-stabilized RNA-seq expression data across all gastroenteropancreatic neuroendocrine tumor samples. Points are colored by tissue of origin (pancreatic neuroendocrine tumors, PanNETs; small-intestinal neuroendocrine tumors, SI-NETs) and shaped by disease status (primary vs metastatic). Samples separate primarily by tissue of origin along the first principal component (PC1, 28% variance), while metastatic status does not produce strong global separation, indicating that tissue identity is the dominant driver of transcriptomic variation across the dataset.

3.2 Tissue-specific transcriptomic structure reveals metastasis-associated variation

To evaluate whether metastatic status contributes to transcriptional variation after accounting for tissue-specific effects, principal component analysis (PCA) was performed separately for pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). By restricting the analysis to a single tissue type, this approach isolates metastasis-associated expression patterns that may be masked in global analyses dominated by tissue identity.

Within the PanNET cohort, PCA revealed partial separation between primary and metastatic samples (Figure 2A). Although substantial overlap was observed, metastatic tumors tended to occupy a broader region of the principal component space, indicating increased transcriptional heterogeneity relative to primary tumors. Several metastatic PanNET samples appeared displaced along the first

principal component, suggesting that metastasis is associated with coordinated, but not uniform, changes in gene expression within this tissue context.

A similar pattern was observed in the SI-NET cohort (Figure 2B). Primary and metastatic SI-NET samples did not form completely distinct clusters; however, metastatic samples again exhibited greater dispersion across principal components. This increased spread suggests that metastatic progression in SI-NETs is associated with variable transcriptional reprogramming rather than a single dominant expression state.

Together, these tissue-specific PCA results indicate that while metastatic status does not drive complete transcriptomic separation within either PanNETs or SI-NETs, it is associated with increased variability and directional shifts in gene expression within each tissue. These observations support the hypothesis that metastasis-related transcriptional changes are context-dependent and motivate downstream differential expression analyses performed independently within each tumor type.

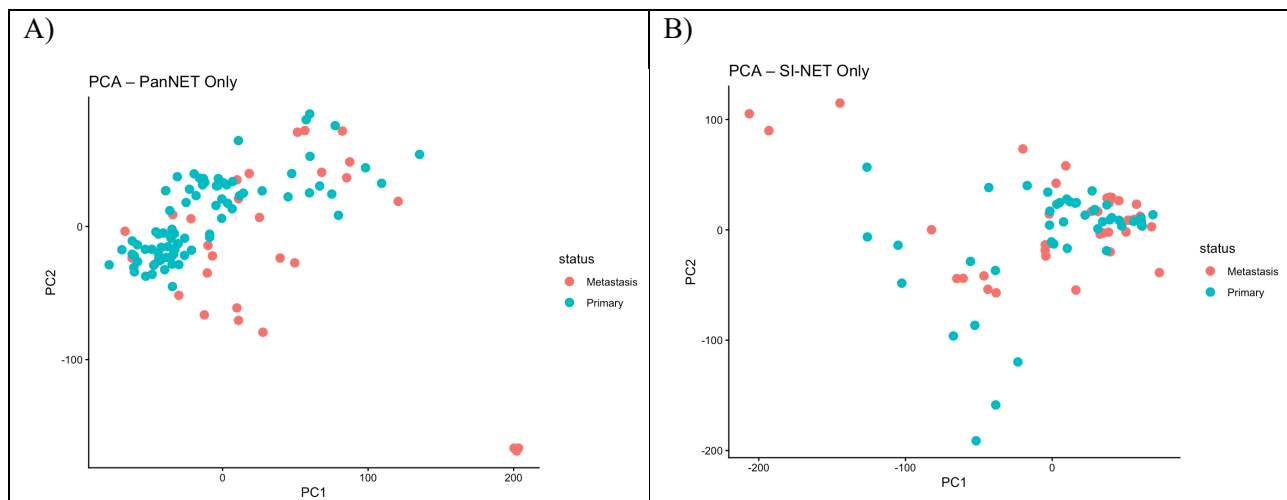


Figure 2. Tissue-specific principal component analysis of gastroenteropancreatic neuroendocrine tumors. Principal component analysis (PCA) of variance-stabilized RNA-seq expression data performed separately within pancreatic neuroendocrine tumors (PanNETs; A) and small-intestinal neuroendocrine tumors (SI-NETs; B). Samples are colored by disease status (primary vs metastatic). In both tissues, primary and metastatic samples show substantial overlap, although metastatic samples exhibit increased dispersion, indicating heterogeneous transcriptional changes associated with metastatic progression within each tissue context.

3.3 Quantitative differential expression analysis reveals extensive metastasis-associated transcriptional reprogramming in PanNETs

After establishing tissue-specific structure and metastasis-associated heterogeneity through principal component analysis, differential expression analysis was performed within pancreatic neuroendocrine tumors (PanNETs) to quantify transcriptomic changes associated with metastatic progression. Restricting this analysis to a single tissue type ensured that observed expression differences reflected metastatic status rather than underlying tissue-of-origin effects.

The PanNET differential expression analysis included 113 samples, consisting of 83 primary tumors and 30 metastatic tumors. Expression differences were evaluated across 35,786 genes with nonzero read counts. Differential expression testing was conducted using DESeq2, which models RNA-seq count data while accounting for biological variability and sequencing depth.

At a false discovery rate threshold of adjusted p -value < 0.05 , a total of 5,305 genes were identified as significantly differentially expressed between metastatic and primary PanNETs (Figure 3A). Among these genes, 4,555 were upregulated in metastatic tumors, while 2,610 were downregulated. This strong imbalance toward upregulated genes indicates that metastatic PanNETs are primarily characterized by the activation of additional gene expression programs, rather than widespread suppression of genes expressed in primary tumors.

Lowly expressed genes were filtered automatically through DESeq2's independent filtering procedure, with 5,569 genes excluded due to insufficient mean expression across samples. Importantly, no genes were identified as outliers based on Cook's distance, indicating that the observed differential expression patterns are not driven by a small number of aberrant samples or extreme expression values. These quality control results support the robustness and reproducibility of the differential expression analysis across the PanNET cohort.

The magnitude of expression changes further underscores the strength of the metastatic transcriptional signal. Many genes exhibited large positive $\log_2$ fold changes, with several showing fourfold to eightfold increases in expression in metastatic tumors relative to primary tumors. This distribution of effect sizes is reflected in the volcano plot, which shows a pronounced right-skewed pattern, indicating that genes upregulated in metastatic tumors dominate the statistically significant signal (Figure 3A). The concentration of genes with both large effect sizes and very small adjusted p -values suggests coordinated transcriptional regulation rather than random gene-level noise.

To assess whether these differentially expressed genes exhibited coordinated expression patterns across samples, a heatmap was generated using the top 40 differentially expressed genes in the PanNET cohort (adjusted p -value < 0.05 ; Figure 3B). Hierarchical clustering revealed that metastatic samples do not form a single uniform transcriptional state. Instead, metastatic tumors segregate into multiple expression patterns, with one subset showing strong coordinated upregulation of metastasis-associated genes, while other metastatic samples remain transcriptionally closer to primary tumors. This structure highlights substantial heterogeneity in metastasis-associated transcriptional programs within PanNETs.

Together, the differential expression results, volcano plot, and heatmap demonstrate that metastatic progression in PanNETs is associated with extensive, structured, and reproducible transcriptional reprogramming affecting a large fraction of the transcriptome. These findings are consistent with tissue-specific PCA results, which showed increased dispersion of metastatic samples relative to primary tumors, and provide a strong quantitative foundation for downstream cross-tissue overlap, pathway enrichment, and signature-based analyses aimed at identifying conserved biological processes underlying metastasis.

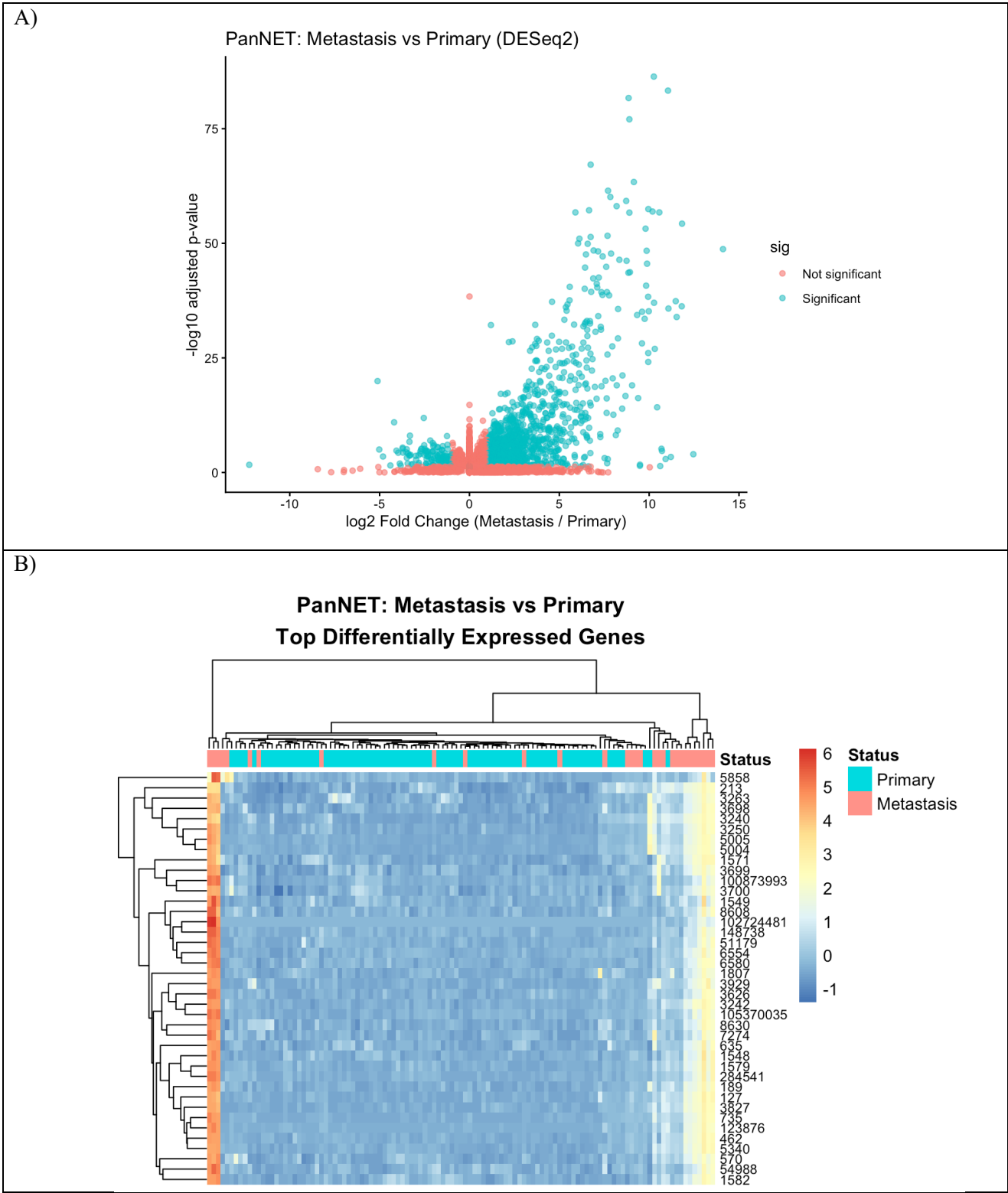


Figure 3. Differential expression analysis reveals extensive metastasis-associated transcriptional reprogramming in PanNETs. (A) Volcano plot showing differential gene expression between metastatic and primary pancreatic neuroendocrine tumors (PanNETs). The x-axis represents log₂ fold change (metastatic vs primary), and the y-axis represents $-\log_{10}$ adjusted p-values. Genes significantly differentially expressed at an adjusted p-value < 0.05 are highlighted, with a pronounced right-skewed distribution indicating widespread upregulation in metastatic tumors. (B) Heatmap of the top 40 differentially expressed genes in PanNETs (adjusted p-value < 0.05). Rows represent genes and columns represent samples. Hierarchical clustering reveals coordinated expression patterns and demonstrated that metastatic tumors segregate into multiple transcriptional states, highlighting heterogeneity in metastasis-associated gene programs.

Metric <chr>	Value <int>
Total genes tested	30236
Significant genes (padj < 0.05)	5305
Upregulated in Metastasis	3546
Upregulated in Primary	1759

Table 1. Summary of differential expression results in pancreatic neuroendocrine tumors (PanNETs). Differential expression analysis was performed using DESeq2 comparing metastatic and primary tumors. Genes with adjusted p-value < 0.05 were considered significant. Upregulated genes are defined based on positive or negative log₂ fold change values relative to the primary tumor reference condition.

3.3 Quantitative differential expression analysis reveals extensive metastasis-associated transcriptional reprogramming in PanNETs

After establishing tissue-specific structure and metastasis-associated heterogeneity through principal component analysis, differential expression analysis was performed within pancreatic neuroendocrine tumors (PanNETs) to quantify transcriptomic changes associated with metastatic progression. Restricting this analysis to a single tissue type ensured that observed expression differences reflected metastatic status rather than underlying tissue-of-origin effects.

The PanNET differential expression analysis included 113 samples, consisting of 83 primary tumors and 30 metastatic tumors. Expression differences were evaluated across 30,236 genes with sufficient read counts for analysis. Differential expression testing was conducted using DESeq2, which models RNA-seq count data while accounting for biological variability and sequencing depth.

At a false discovery rate threshold of adjusted p-value < 0.05, a total of 5,305 genes were identified as significantly differentially expressed between metastatic and primary PanNETs (Figure 3A; Table 1). This represents approximately 17.5% of all tested genes, indicating that metastatic progression is associated with widespread transcriptomic changes affecting a substantial portion of the genome.

Among these significant genes, 3,546 were upregulated in metastatic tumors, while 1,759 were upregulated in primary tumors. This strong imbalance toward upregulation in metastatic samples indicates that metastatic PanNETs are primarily characterized by activation of additional gene expression programs rather than widespread repression of primary tumor-associated genes. This pattern suggests that metastasis involves the acquisition of new transcriptional states rather than simple loss of existing cellular identity.

Lowly expressed genes were filtered automatically through DESeq2's independent filtering procedure, with genes of insufficient expression excluded prior to statistical testing. Importantly, no genes were identified as outliers based on Cook's distance, indicating that the observed differential expression patterns are not driven by a small number of aberrant samples or extreme expression values. These quality control results support the robustness and reproducibility of the differential expression analysis across the PanNET cohort.

The magnitude of expression changes further underscores the strength of the metastatic transcriptional signal. Many genes exhibited large positive log₂ fold changes, with several showing fourfold to eightfold increases in expression in metastatic tumors relative to primary tumors. This distribution of effect sizes is reflected in the volcano plot, which shows a pronounced right-skewed pattern, indicating that genes upregulated in metastatic tumors dominate the statistically significant signal (Figure 3A). The concentration of genes with both large effect sizes and very small adjusted p-values suggests coordinated transcriptional regulation rather than random gene-level noise.

To assess whether these differentially expressed genes exhibited coordinated expression patterns across samples, a heatmap was generated using the top 40 differentially expressed genes in the PanNET cohort (adjusted p-value < 0.05; Figure 3B). Rows represent genes and columns represent samples, with expression values standardized across genes to highlight relative differences. Hierarchical clustering revealed that metastatic samples do not form a single uniform transcriptional state. Instead, metastatic tumors segregate into multiple expression patterns, with one subset showing strong coordinated upregulation of metastasis-associated genes, while other metastatic samples remain transcriptionally closer to primary tumors. This structure highlights substantial heterogeneity in metastasis-associated transcriptional programs within PanNETs.

Together, the differential expression results, volcano plot, and heatmap demonstrate that metastatic progression in PanNETs is associated with extensive, structured, and reproducible transcriptional reprogramming affecting a large

fraction of the transcriptome. These findings are consistent with tissue-specific PCA results, which showed increased dispersion of metastatic samples relative to primary tumors, and provide a strong quantitative foundation for downstream cross-tissue overlap, pathway enrichment, and signature-based analyses aimed at identifying conserved biological processes underlying metastasis.

3.4 Differential expression analysis reveals extensive metastasis-associated transcriptional changes in SI-NETs

To determine whether the extensive transcriptional reprogramming observed in pancreatic neuroendocrine tumors (PanNETs) is also present in small-intestinal neuroendocrine tumors (SI-NETs), differential expression analysis was performed comparing metastatic tumors to primary tumors within the SI-NET cohort.

Differential expression testing identified 4,053 genes as significantly differentially expressed at an adjusted p-value threshold of < 0.05 (Table 2). Compared to PanNETs, where metastasis was associated with widespread and strongly skewed transcriptional changes, SI-NETs exhibited a more moderate and balanced differential expression profile. Both upregulated and downregulated genes were observed, with a less pronounced bias toward upregulation in metastatic samples.

The distribution of effect sizes in SI-NETs was also more symmetric, with fewer genes showing extremely large log₂ fold changes. This pattern suggests that metastatic progression in SI-NETs involves more subtle and distributed transcriptional changes rather than large-scale activation of gene expression programs. These findings are consistent with tissue-specific PCA results, which showed substantial overlap between primary and metastatic SI-NET samples and greater intrinsic heterogeneity across the cohort.

Overall, while metastasis in SI-NETs is associated with statistically significant gene expression changes, the magnitude and structure of these changes are less dramatic than those observed in PanNETs. This contrast highlights the importance of tissue context in shaping the transcriptional impact of metastatic progression and reinforces the need for cross-tissue comparative analysis to identify conserved biological mechanisms.

Metric <chr>	Value <dbl>
PanNET significant genes	5305.00
SI-NET significant genes	4053.00
Shared significant genes	1308.00
Percent of PanNET shared	24.66
Percent of SI-NET shared	32.27
Shared genes (same direction)	1139.00
Shared genes (opposite direction)	169.00

Table 2. Cross-tissue comparison of differential expression results between pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). Significant genes (adjusted p-value < 0.05) were identified independently within each tissue. Shared genes represent the intersection of significant genes between tissues. Directionality reflects whether genes are regulated in the same or opposite direction in metastatic tumors relative to primary tumors.

3.5 Cross-tissue comparison reveals a conserved metastasis-associated transcriptional program

To determine whether metastasis-associated transcriptional changes are shared across tumor types, differential expression results from pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs) were compared. Significant genes identified independently within each tissue (adjusted p-value < 0.05) were intersected to identify a conserved set of metastasis-associated genes.

This comparison revealed a substantial overlap between the two tumor types. A total of 1,308 genes were significantly differentially expressed in both PanNETs and SI-NETs (Table 2). This represents approximately 24.7% of PanNET-associated genes and 32.3% of SI-NET-associated genes, indicating that a meaningful fraction of metastasis-associated transcriptional changes are conserved across tissues despite differences in tissue of origin and overall transcriptional structure.

To further assess the consistency of these shared genes, the direction of differential expression was examined by comparing log₂ fold changes between tissues. The majority of shared genes (1,139 genes; approximately 87%) exhibited consistent

directionality, meaning that they were either upregulated in metastatic tumors in both tissues or downregulated in both tissues. In contrast, only 169 shared genes showed opposite directions of regulation between PanNETs and SI-NETs.

To visualize this relationship, a scatter plot of log₂ fold changes for shared genes was generated, comparing expression changes in PanNETs and SI-NETs (Figure 4). Each point represents a gene shared between the two tissue-specific analyses. The majority of genes fall along the diagonal, indicating consistent direction and magnitude of change across tissues. This pattern demonstrates that metastasis-associated transcriptional changes are not only shared but are also coordinated in a similar direction across tumor types.

Together, these results provide strong evidence for the existence of a conserved metastasis-associated transcriptional program in gastroenteropancreatic neuroendocrine tumors. While tissue-specific differences influence the magnitude and structure of gene expression changes, a core set of genes appears to be consistently associated with metastatic progression across both pancreatic and small-intestinal neuroendocrine tumors.

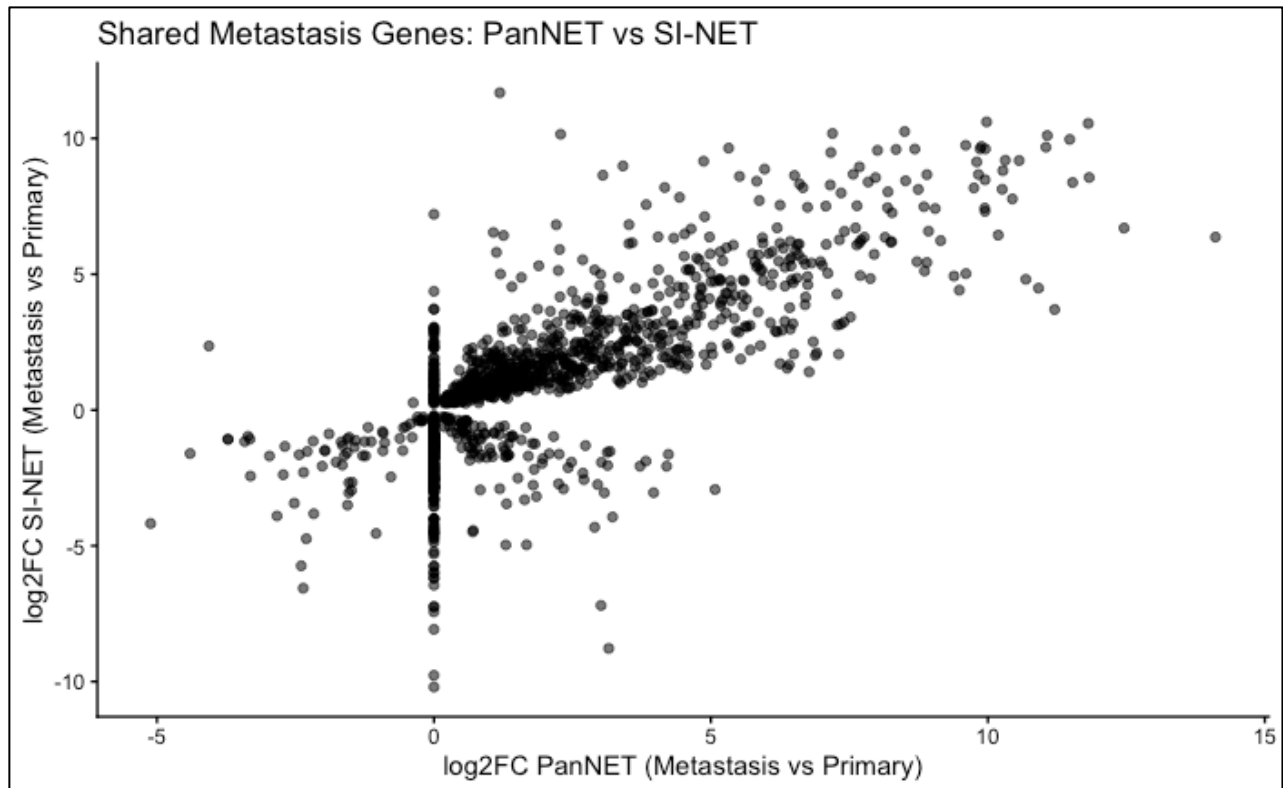


Figure 4. Cross-tissue comparison of metastasis-associated gene expression changes in PanNETs and SI-NETs. Scatter plot of log₂ fold changes for genes significantly differentially expressed in both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). Each point represents a shared gene. The x-axis shows log₂ fold change (metastasis vs primary) in PanNETs, and the y-axis shows log₂ fold change in SI-NETs. The majority of genes cluster along the diagonal, indicating consistent directionality of expression changes across tissues and supporting the presence of conserved metastasis-associated transcriptional program.

3.6 Shared metastasis-associated genes are enriched for metabolic processes

To characterize the biological functions associated with the conserved metastasis-associated gene set, pathway enrichment analysis was performed on genes that were significantly upregulated in metastatic tumors in both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). This analysis focused on a subset of 887 genes that exhibited consistent upregulation in metastatic samples across both tissues, representing a core shared transcriptional program.

Gene Ontology (GO) biological process enrichment analysis revealed strong and coherent enrichment

for metabolic processes (Figure 5). Among the most significantly enriched pathways were those related to amino acid metabolism, organic acid metabolism, and small molecule metabolic processes. These results indicate that metastatic tumors exhibit coordinated upregulation of metabolic pathways that may support increased energy demand, biosynthetic activity, and adaptation to new microenvironments.

The enrichment of metabolic pathways suggests that metastatic progression in neuroendocrine tumors is associated with systematic metabolic reprogramming rather than isolated gene-level changes. Such reprogramming is a well-established hallmark of cancer, enabling tumor cells to sustain

proliferation, survive under stress conditions, and successfully colonize distant tissues.

Importantly, the enrichment results were highly consistent across multiple related biological processes, with overlapping gene sets contributing to several top-ranked pathways. This pattern reinforces the conclusion that the observed signal reflects coordinated activation of functional gene networks rather than stochastic variation.

Overall, pathway enrichment analysis demonstrates that the shared metastasis-associated transcriptional program is strongly enriched for metabolic processes, highlighting metabolic adaptation as a central and conserved feature of metastatic progression in gastroenteropancreatic neuroendocrine tumors.

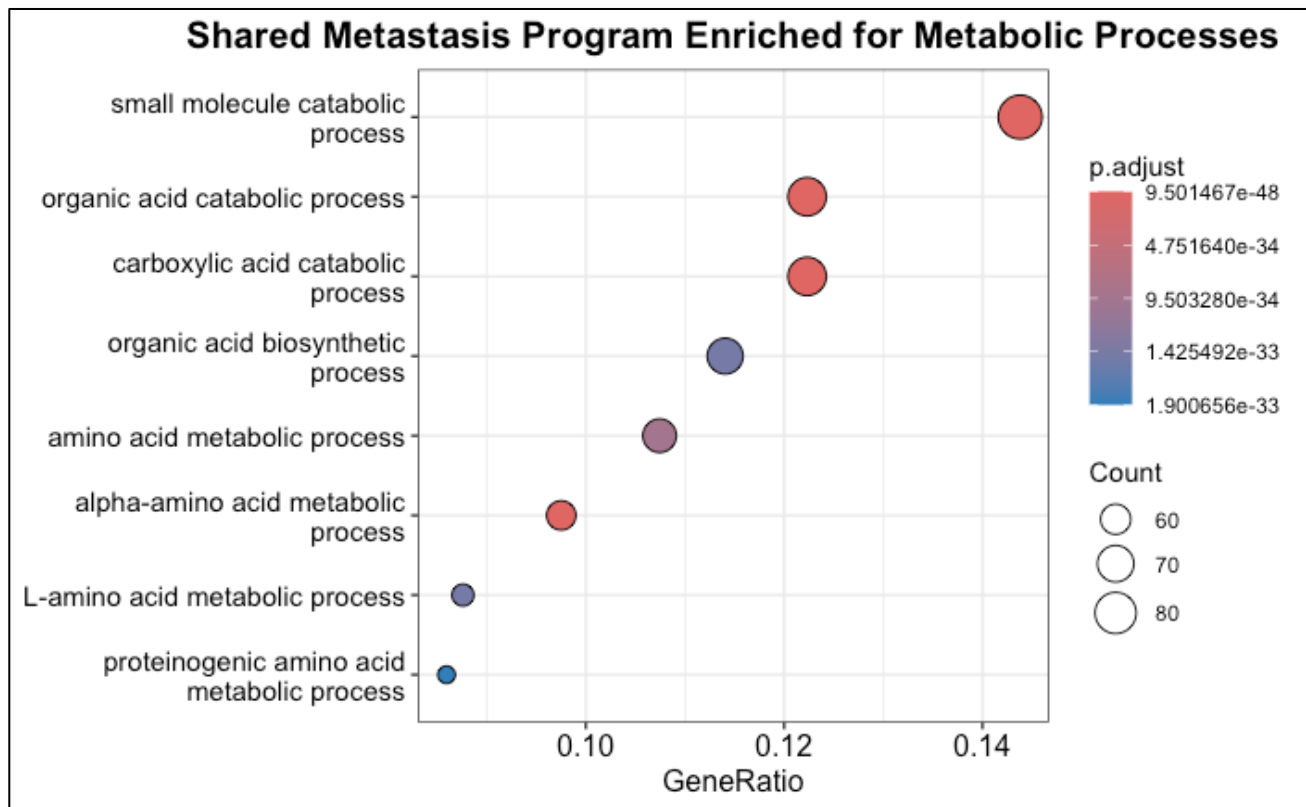


Figure 5. Gene Ontology enrichment analysis of shared metastasis-associated genes. Dot plot showing enriched biological processes among genes consistently upregulated in metastatic tumors in both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). Each point represents a Gene Ontology (GO) term, with point size corresponding to the number of genes associated with the term and color indicating statistical significance. Enrichment results highlight strong overrepresentation of metabolic processes, including amino acid, organic acid, and small molecule metabolism, indicating coordinated metabolic reprogramming associated with metastatic progression.

3.7 A shared metastasis gene signature is elevated in metastatic tumors across tissues

To validate the activity of the conserved metastasis-associated transcriptional program at the sample level, a shared gene signature was constructed using genes consistently upregulated in metastatic tumors in both pancreatic neuroendocrine tumors

(PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). This signature consisted of 887 genes, representing a robust and directionally consistent subset of the shared metastasis-associated gene set identified through cross-tissue comparison.

For each sample, a signature score was calculated by averaging z-score normalized expression values across all genes in the shared signature. This approach quantifies the overall activation of the metastasis-associated transcriptional program within individual tumors while controlling for differences in baseline expression across genes.

Comparison of signature scores between primary and metastatic tumors revealed a clear and statistically significant increase in metastatic samples (Figure 6). Metastatic tumors exhibited higher signature scores than primary tumors, indicating stronger activation of the shared transcriptional program. This difference was highly significant (Welch's two-sample t-test, $p = 2.4 \times 10^{-6}$), demonstrating that the observed transcriptional pattern is not due to random variation.

When stratified by tissue type, this pattern remained consistent. Both PanNET and SI-NET metastatic samples showed elevated signature scores relative to their respective primary tumors, confirming that the shared metastasis-associated program is active across both anatomical contexts. Although variability was observed within each group, the overall shift toward higher scores in metastatic samples was consistent and reproducible.

These results provide independent validation of the conserved metastasis-associated transcriptional program identified through differential expression and pathway enrichment analyses. By demonstrating that this program is consistently elevated at the sample level, the signature-based approach confirms that shared gene expression changes reflect coordinated biological processes rather than isolated statistical findings.

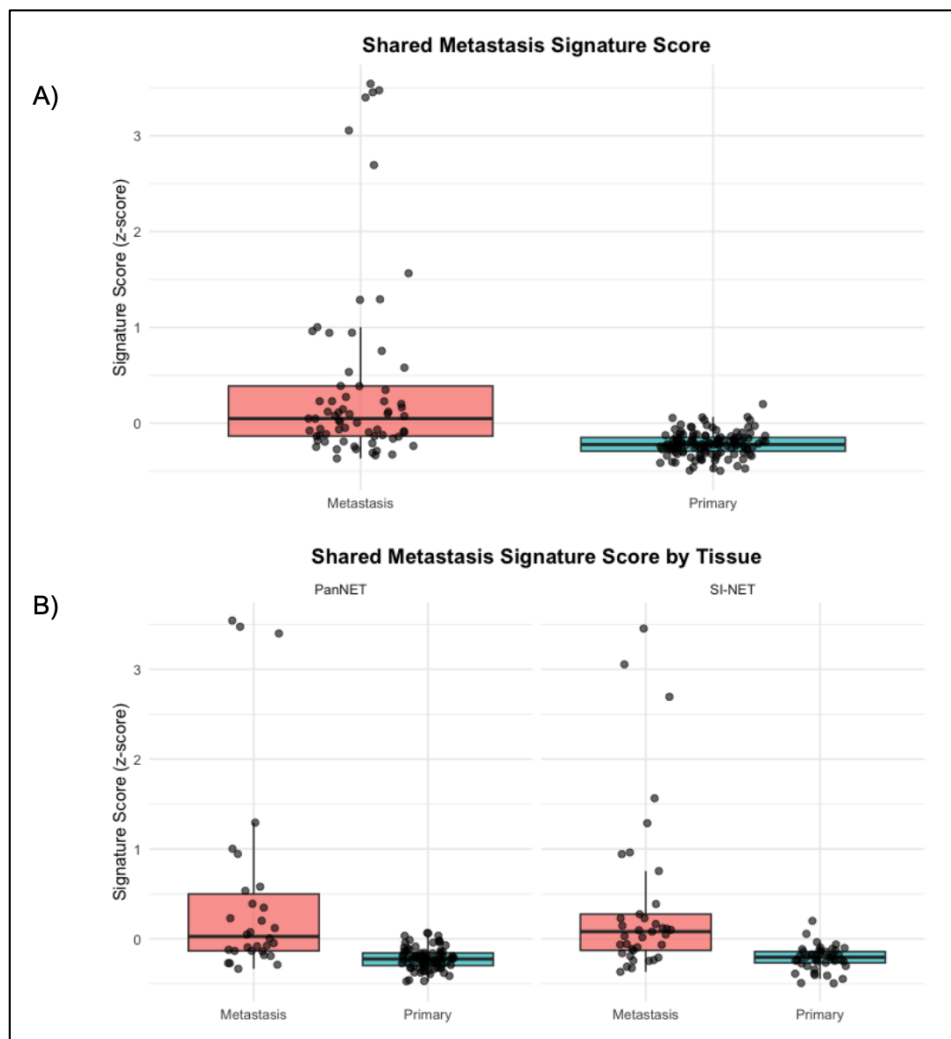


Figure 6. Shared metastasis gene signature is elevated in metastatic tumors across tissues. (A) Boxplot showing signature scores calculated from genes consistently upregulated in metastatic tumors in both pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). Each point represents an individual sample. Metastatic tumors exhibit significantly higher signature scores than primary tumors (Welch's t-test, $p = 2.4 \times 10^{-6}$). (B) Stratified analysis by tissue type demonstrates that this pattern is consistent in both PanNET and SI-NET cohorts, indicating that the shared metastasis-associated transcriptional program is active across anatomical contexts.

4. Conclusion

This project used RNA sequencing (RNA-seq) data to understand how gene expression changes when neuroendocrine tumors become metastatic.

Specifically, I analyzed tumors from two different tissue types: pancreatic neuroendocrine tumors (PanNETs) and small-intestinal neuroendocrine tumors (SI-NETs). The main goal was to determine whether metastasis behaves differently in each tissue or whether there are shared biological patterns across both tumor types.

First, I examined the overall structure of the data using principal component analysis (PCA). This showed that the strongest differences in gene expression are driven by tissue of origin, meaning that pancreatic and small-intestinal tumors are fundamentally different at the transcriptional level. However, metastatic tumors also showed additional variation within each tissue, suggesting that metastasis introduces its own changes on top of the underlying tissue identity.

Next, I performed differential expression analysis to identify genes that change between primary and metastatic tumors. In PanNETs, metastasis was associated with large-scale transcriptional changes, with over 5,000 genes significantly differentially expressed. In contrast, SI-NETs showed more moderate changes, indicating that the impact of metastasis is not identical across tissues. This demonstrated that metastasis is a tissue-dependent process, with different magnitudes of gene expression changes depending on the tumor type.

Despite these differences, I then asked a key question: are there any gene expression changes that are shared between both tumor types? By comparing results across tissues, we identified 1,308 genes that were significantly different in both PanNETs and SI-NETs. Importantly, most of these genes (about 87%) changed in the same direction in both tissues, meaning they were either consistently increased or decreased in metastatic tumors. This

showed that, even though the overall behavior of metastasis differs by tissue, there is a core set of genes that behave similarly across both tumor types.

To understand what these shared genes do, I performed pathway enrichment analysis. This revealed that many of the shared genes are involved in metabolic processes, such as amino acid metabolism and small molecule metabolism. This suggests that metastatic tumors may rely on changes in metabolism to support their growth and survival, which is consistent with known principles of cancer biology.

Finally, I tested whether this shared set of genes is actually active at the level of individual tumor samples. To do this, I created a gene signature using genes that were consistently upregulated in metastatic tumors across both tissues. I then calculated a signature score for each sample. This analysis showed that metastatic tumors had significantly higher signature scores than primary tumors, and this pattern was consistent in both PanNETs and SI-NETs. This confirms that the shared gene program is not just statistically significant, but is actively expressed in metastatic tumors.

Overall, this project shows that metastasis in neuroendocrine tumors involves both tissue-specific changes and a shared underlying biological program. While pancreatic tumors undergo more dramatic transcriptional changes and small-intestinal tumors show more subtle shifts, both tumor types activate a common set of genes related to metabolism during metastasis. This suggests that metabolic reprogramming may be a fundamental feature of metastatic progression across neuroendocrine tumors.

In summary, this analysis demonstrates that combining differential expression, cross-tissue comparison, pathway analysis, and sample-level validation provides a powerful way to identify

meaningful biological patterns in complex cancer data. These findings highlight the importance of looking beyond individual genes to understand broader transcriptional programs and provide a foundation for future studies investigating shared mechanisms of cancer metastasis.

5. References

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